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Interactivity as a peripheral cue in the online persuasion process for low-involvement products

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Abstract

Purpose – This study examines the role of website interactivity in shaping consumer responses in the context of low-involvement products. Drawing on the Elaboration Likelihood Model it investigates whether interactivity functions primarily as a peripheral cue under low involvement and whether increasing levels of interactivity enhance persuasion in a linear or non-linear manner.

Design/methodology/approach – An experimental design was employed using three versions of a fictitious refreshment soft drink brand website, representing low, medium and high levels of interactivity. Participants were randomly assigned to one condition and completed measures of perceived interactivity, cognitive responses using a thought-listing technique, attitudes toward the website and the brand, intention to revisit the website and pre-purchase intention. Attitude strength was assessed one week later without re-exposure to the website.

Findings – The findings indicate that website interactivity plays a meaningful but non-linear role in persuasion. The absence of a linear progression highlights the importance of identifying optimal rather than maximal interactivity, particularly in low-involvement contexts. Medium interactivity generated the highest level of product-oriented cognitive responses, whereas website-oriented thoughts were not significantly affected by interactivity level. Attitude strength was significantly higher under high interactivity compared with medium and low interactivity. Although attitudes and behavioural intentions followed the expected directional pattern, most behavioural effects were not statistically significant.

Research limitations/implications – The findings highlight important implications for theory and practice by showing that website interactivity operates conditionally as a peripheral cue in low-involvement contexts and produces non-linear cognitive and attitudinal outcomes. This underscores the need for level-specific interactivity design and more context-sensitive theoretical models of digital persuasion grounded in the Elaboration Likelihood Model. At the same time, the study is limited by its focus on a single low-involvement product category, one experimental setting, and reliance on self-reported attitudinal and behavioural measures, which may restrict generalisability to other products, platforms or real purchase behaviour. Future research should examine high-involvement contexts, longitudinal effects, and more diverse user populations.

Practical implications – The findings offer several practical implications for managers designing digital marketing strategies and brand websites. First, the results caution against the assumption that more interactivity is always better. For low-involvement products, increasing interactivity does not necessarily lead to proportional gains in engagement or behavioural intentions. Managers should therefore avoid overloading websites with interactive features that increase cognitive effort without delivering clear value. Second, the finding that low-interactivity websites generate more positive website attitudes highlights the importance of simplicity and usability. In low-involvement contexts, consumers may prioritize efficiency, clarity, and ease of navigation over rich interactive experiences. Streamlined website designs that minimize friction may therefore be more effective. Third, high levels of interactivity appear to enhance attitude strength, suggesting that immersive experiences may still be valuable when the strategic objective is to build favourable attitudes rather than to drive immediate purchase behaviour. Interactive elements that foster enjoyment, perceived control, or engagement may strengthen affective associations with the product or brand. Finally, the absence of significant effects on behavioural intentions underscores the need for integrated digital strategies. Interactivity alone may be insufficient to drive action in low-involvement contexts. Managers should combine interactive website features with complementary tactics, such as promotions, reminders, or repeated exposure across channels, to reinforce behavioural responses over time.

Social implications – The study highlights ethical considerations in digital persuasion, showing that website interactivity may shape consumer attitudes through peripheral processing in low-involvement contexts.



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Designing simpler, more transparent digital environments may therefore better support informed choice, consumer autonomy, and digital well-being.

Originality/value – By explicitly testing three levels of website interactivity in a low-involvement context, this study clarifies that increased interactivity does not always uniformly enhance persuasion. The findings extend ELM-based explanations of online persuasion by demonstrating the contingent, context-dependent and sometimes diminishing effects of interactivity in digital brand environments.

Keywords Website interactivity, Online persuasion, Low-involvement product, Elaboration likelihood model, ELM, Thought listing, Attitude strength

Paper type Research article

1. Introduction

During 2025, the total number of internet users was estimated at 5.56 billion, indicating that nearly two out of every three people worldwide had access to the internet (Statista, 2025). This widespread diffusion of digital technologies has fundamentally reshaped firm–consumer communication, shifting traditional one-way models toward more dynamic and interactive exchanges. As digitalization has transformed how companies interact with consumers, strategic digital transformation has become a prerequisite for organisational survival and growth rather than a source of competitive advantage (Khoshroo and Talari, 2025). In this environment, firms increasingly rely on interactive marketing techniques to enhance customer experiences and leverage technological advancements across the entire shopping journey (Hood *et al.*, 2015; Du *et al.*, 2022; Konstantoulaki *et al.*, 2025). At the same time, as consumers' familiarity with digital technologies has increased, their expectations regarding responsiveness, engagement, and control have intensified (Lütjens *et al.*, 2022). Recent research further highlights that digital marketing is evolving toward dynamic, technology-mediated customer journeys, where interactive touchpoints increasingly shape consumer responses (Gensler and Rangaswamy, 2025).

Within this evolving landscape, the official brand website remains central to firm–consumer communication, serving as a primary source of information for internet users. Well-designed websites facilitate interactive communication, support long-term relationship development, and contribute to consumer engagement and loyalty (Voorveld *et al.*, 2013; Liao *et al.*, 2019; Quan *et al.*, 2020; Jongmans *et al.*, 2022). Website interactivity, in particular, has long been regarded as a key determinant of online communication effectiveness (Ghose and Dou, 1998; Macias, 2003; Song and Zinkhan, 2008; Hari *et al.*, 2022). However, the mere addition of interactive elements does not guarantee effective communication. Interactivity is not a dichotomous attribute; rather, websites vary in both the level and type of interactivity they provide, enabling users to influence content and form to different degrees (Coyle and Thorson, 2001; Lombard and Snyder-Duch, 2001; Coursaris and Sung, 2012; Fazli-Salehi *et al.*, 2022). Consequently, different levels of interactivity may lead to distinct communication and persuasion outcomes.

Despite extensive research, empirical findings regarding the effectiveness of website interactivity remain mixed. While some studies document positive effects on engagement, attitudes, and evaluations (Devaraj *et al.*, 2002; McMillan *et al.*, 2003; Thorson and Rodgers, 2006; Yim *et al.*, 2017), others report neutral or even negative effects, particularly when interactivity increases cognitive load or distracts users from core message content (Ariely, 2000; Fortin and Dholakia, 2005; Sohn *et al.*, 2007). More recent research suggests that the effects of interactivity may be non-linear, with moderate levels sometimes proving more effective than very high levels (Mai *et al.*, 2014; Hood *et al.*, 2015; Yang and Shen, 2018; Wu, 2019; Maduku *et al.*, 2024).

To account for these inconsistencies, scholars increasingly emphasize the role of moderating factors, particularly product involvement (Liu and Shrum, 2009; Li and Shi, 2025; Shabani Shojaei and Barbosa, 2026). According to the Elaboration Likelihood Model (ELM), involvement determines whether individuals follow the central or the peripheral route of information processing (Petty and Cacioppo, 1981, 1986). Under low-involvement

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conditions, persuasion is more likely to be driven by peripheral cues rather than by extensive cognitive elaboration of message arguments.

Although recent research indicates that interactivity can enhance engagement in low-involvement contexts, its effects on thought elicitation, attitude strength, and behavioural intentions appear subtle and highly contingent on design and perception (Islam *et al*., 2021; Joo and Yang, 2023; Pangarkar and Kirk, 2025). As a result, the persuasive role of different levels of website interactivity for low-involvement products remains insufficiently understood. Emerging research on AI-driven interactive environments suggests that increasingly complex and personalized digital interactions can shape customer experience in multifaceted ways, sometimes enhancing and sometimes complicating consumer responses (Hardcastle *et al*., 2025; Desveaud *et al*., 2024).

Against this backdrop, the present study aims to address the following research objectives: (1) to examine how varying levels of website interactivity influence cognitive responses in the context of a low-involvement product, (2) to investigate the effect of interactivity on attitude strength, (3) to assess its impact on attitudes toward the website and the brand as well as behavioural intentions, and (4) to determine whether these effects follow a linear or non-linear pattern. Using an experimental design, three versions of a brand website - characterized by low, medium, and high perceived interactivity - were developed. The study investigates how these interactivity levels affect information processing related to the website and the product, attitude strength, attitudes toward the website and the brand, and pre-purchase behaviours, including intentions to revisit the website and to purchase the product online. Drawing on the Elaboration Likelihood Model (ELM), website interactivity is conceptualised as a peripheral cue whose persuasive influence may vary depending on its intensity under low-involvement conditions (Petty and Wegener, 1998).

2. Literature review

2.1 Elaboration likelihood model (ELM) and online persuasion

The present study is grounded in the ELM, which serves as the primary theoretical framework guiding the conceptual model, the selection of variables, and the formulation of hypotheses. ELM provides a well-established basis for understanding how consumers process persuasive information in digital environments and is particularly relevant for examining the role of contextual cues such as website interactivity under low-involvement conditions.

The ELM explains persuasion as a function of different levels of cognitive elaboration (Petty and Cacioppo, 1986). It distinguishes between a central route, involving systematic evaluation of message arguments, and a peripheral route, where attitudes are shaped by cues requiring minimal cognitive effort. Importantly, ELM conceptualises persuasion along an elaboration continuum rather than as a strict dichotomy (Petty *et al*., 2009).

Attitudes formed through high elaboration are more stable, resistant to counter-persuasion, and predictive of behaviour than those formed under low elaboration (Krosnick and Petty, 1995; Wagner and Petty, 2022). However, when motivation or ability to process information is low, peripheral cues related to the communication context may dominate attitude formation (Petty and Cacioppo, 1986). As digital environments have become increasingly interactive, scholars have applied ELM to online persuasion, highlighting how interactivity complicates traditional distinctions between central and peripheral processing (Liu and Shrum, 2002).

Within ELM, the third (3rd) postulate suggests that the same variable may influence persuasion by serving as an argument, acting as a peripheral cue, or affecting elaboration, depending on contextual conditions (Petty and Wegener, 1998). The sixth (6th) postulate further posits that when both motivation and ability are low, attitude change is driven primarily by peripheral cues. These principles are particularly relevant for understanding website interactivity, which may either facilitate or inhibit persuasion depending on users' experience and involvement (Liu and Shrum, 2009).

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2.2 Product involvement as a boundary condition

Product involvement reflects the degree of personal relevance consumers assign to a product and plays a central role in shaping information processing (Petty and Cacioppo, 1986). High-involvement products motivate systematic evaluation of arguments, whereas low-involvement products are more likely to be processed using heuristic or affective cues (Park and Moon, 2003; Schumann *et al*., 2012).

In online contexts, involvement influences browsing behaviour, interaction intensity, and responsiveness to interactive features (Mathwick and Rigdon, 2004; Levy and Nebenzahl, 2008). For high-involvement products, interactivity can enhance elaboration and lead to stronger attitudes and more informed decisions (Palla *et al*., 2013). For low-involvement products, however, interactivity tends to operate through peripheral mechanisms and yields more limited persuasive effects (Lee, 2020). Recent studies confirm that involvement moderates the impact of interactive website features on consumer responses (Chen and Zhu, 2022; Pangarkar and Kirk, 2025; Palla *et al*., 2023).

2.3 Website interactivity: levels and perceptions

Interactivity has been defined as the extent to which users can participate in modifying the form and content of a mediated environment in real time (Steuer, 1992). Research distinguishes between actual interactivity, referring to objectively embedded features, and perceived interactivity, which captures users' subjective experience of control, responsiveness, and two-way communication (McMillan *et al*., 2003; Wu, 2006; Voorveld *et al*., 2011).

Actual interactivity does not necessarily translate into perceived interactivity. Websites with numerous interactive features may still be experienced as static if users do not engage with them, while perceived interactivity may be high even when objective interactive structures are limited (McMillan *et al*., 2003; Voorveld *et al*., 2018). Consequently, perceived interactivity has been identified as a more proximal determinant of engagement and persuasion (Joo and Yang, 2023; Xiang and Chae, 2022; Konstantoulaki *et al*., 2025). Prior research further suggests that perceived interactivity comprises active user control, two-way communication, and synchronicity (Blattberg and Deighton, 1991; Novak *et al*., 2000).

2.4 Conflicting and non-linear effects of interactivity

Although many studies document positive effects of website interactivity, others highlight its potential drawbacks. Excessive interactivity may increase cognitive load, fragment attention, and reduce message comprehension, particularly when motivation or ability to process information is low (Ariely, 2000; Fortin and Dholakia, 2005; Sohn *et al*., 2007). Meta-analytic and empirical evidence suggests that interactivity effects are often non-linear, with moderate levels outperforming very high levels in certain contexts (Yang and Shen, 2018; Mai *et al*., 2014; Wu, 2019; Maduku *et al*., 2024).

Recent meta-analytic evidence suggests that the effects of interactivity on consumer responses are contingent and vary across outcome types and contextual conditions, rather than being uniformly positive (Fan *et al*., 2025). Consistent with this perspective, meta-analytic findings on digital persuasion further indicate that personalization and other interactive cues do not exert uniform effects but instead shape attitudes and behavioural outcomes in nuanced and context-dependent ways (Yeo *et al*., 2025). Collectively, these findings reinforce the view that interactive features may produce diminishing or even counterproductive effects when their intensity exceeds users' cognitive processing capacity. Overall, the literature indicates that website interactivity is a multifaceted and context-dependent construct. In low-involvement settings, interactivity is likely to function primarily as a peripheral cue, shaping attitudes through experiential and affective mechanisms rather than through systematic elaboration. At the same time, its effects are not uniformly positive, but depend on the level and design of interactivity, with potential diminishing or counterproductive outcomes when cognitive processing capacity is exceeded.

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3. Hypotheses development

Drawing on the ELM, the present study conceptualises website interactivity as a contextual feature that may influence persuasion through different cognitive mechanisms depending on users' motivation and ability to process information (Petty and Cacioppo, 1986; Li and Shi, 2025). In low-involvement contexts, individuals are less motivated to engage in systematic evaluation of message arguments and are therefore more likely to rely on peripheral cues. Under such conditions, characteristics of the communication medium itself - such as website interactivity - may play a decisive role in shaping cognitive, attitudinal, and behavioural responses.

According to ELM's sixth (6th) postulate, when both motivation and ability to process information are low, attitude change is primarily driven by peripheral cues rather than by extensive elaboration of message arguments (Petty and Wegener, 1998). At the same time, ELM's third (3rd) postulate suggests that the same variable may influence persuasion in different ways -either facilitating or inhibiting elaboration - depending on situational factors (Petty and Cacioppo, 1986). This theoretical logic implies that website interactivity may function as a peripheral cue in low-involvement settings, while its effectiveness may vary depending on the level of interactivity employed.

Prior research supports this contingent view. Studies indicate that interactivity can stimulate engagement, enjoyment, and information processing (Sicilia *et al*., 2005; Oh and Sundar, 2015), yet excessive interactivity may increase cognitive load and distract users, thereby suppressing persuasion (Ariely, 2000; Fortin and Dholakia, 2005; Sohn *et al*., 2007). Liu and Shrum (2009) further demonstrate that interactivity may facilitate elaboration for highly involved or experienced users, while inhibiting cognitive processing for users with low motivation or limited ability. These findings suggest that the cognitive impact of interactivity is likely to differ across levels of interactivity, particularly in low-involvement product contexts.

3.1 Thought elicitation

For low-involvement products, persuasion is often driven by experiential and contextual cues rather than by systematic evaluation of message arguments (Petty and Cacioppo, 1986). Moderate levels of interactivity may encourage exploration and engagement, thereby increasing cognitive responses related to both the website and the product. In contrast, very high levels of interactivity may overwhelm users and reduce effective information processing. Accordingly, the extent of thought elicitation is expected to vary across different levels of website interactivity.

From an ELM perspective, when involvement is low, individuals rely on peripheral cues that shape attention and cognitive engagement without necessarily triggering systematic processing. As an experiential feature of the communication medium, interactivity may influence both the quantity and the focus of cognitive responses, depending on its intensity.

H1. When individuals are exposed to a website promoting a low-involvement product, the number of elicited thoughts related to

(1) the website and

(2) the product will differ across levels of interactivity:

   a. from low to medium,

   b. from medium to high, and

   c. from low to high.

3.2 Attitude strength

Attitude strength reflects the certainty, accessibility, and durability of attitudes and is closely linked to future behaviour (Krosnick and Petty, 1995; DeMarree *et al*., 2007). Although attitudes formed under low involvement are typically weaker than those formed through

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central processing, peripheral cues may still enhance attitude strength by increasing confidence, perceived knowledge, or affective engagement. In digital environments, website interactivity has been shown to foster immersion and perceived control, which may reinforce attitude accessibility even when elaboration is limited (Flavián *et al*., 2019).

Empirical evidence suggests that interactivity is positively associated with attitude formation and evaluation, though these effects may not be linear (Devaraj *et al*., 2002; Yim *et al*., 2017; Yang and Shen, 2018). Therefore, attitude strength is expected to differ across levels of website interactivity. It is expected that as interactivity increases the formulated attitudes will become stronger.

Peripheral cues may also influence attitude strength by enhancing confidence, accessibility, and affective reinforcement, even in the absence of extensive elaboration. Higher levels of interactivity may provide stronger experiential signals, potentially leading to more durable and accessible attitudes.

H2. When individuals are exposed to a website promoting a low-involvement product, attitude strength will become stronger as the levels of interactivity becomes:

a. from low to medium,

b. from medium to high, and

c. from low to high.

3.3 Attitudes and behavioural intentions

Website interactivity has also been linked to behavioural intentions such as intentions to revisit a website and to purchase a product online. Under low-involvement conditions, interactivity may operate primarily as a peripheral cue that enhances affective responses and overall evaluations without extensive cognitive deliberation (Liu and Shrum, 2009). Prior studies indicate that perceived interactivity increases immersion and value gratification, which in turn foster positive attitudes toward websites and brands and strengthen behavioural intentions (Islam *et al*., 2021; Joo and Yang, 2023).

However, consistent with ELM's contingent logic, excessive interactivity may distract users and weaken the translation of positive attitudes into behavioural intentions (Ariely, 2000; Fortin and Dholakia, 2005). Moreover, it is generally accepted that attitude affects pre-purchase behavior (Koc *et al.*, 2025; Saeed and Mohy-Ud-Din, 2026). Accordingly, behavioural outcomes are expected to vary depending on the level of interactivity employed.

Although peripheral cues can shape attitudes, their translation into behavioural intentions is not always direct. Under low-involvement conditions, limited motivation may weaken the relationship between attitudes and behavioural outcomes, suggesting that the effects of interactivity on behaviour may be less pronounced.

H3. Compared to websites with low or medium interactivity, exposure to a highly interactive website promoting a low-involvement product will lead to:

(1) more positive attitudes toward the website and the brand,

(2) stronger intentions to revisit the website, and

(3) more favourable pre-purchase intentions.

4. Method

4.1 Research design

To examine the effects of website interactivity on consumer responses in a low-involvement product context, an experimental research design was employed. Experimental methods are particularly appropriate for investigating causal relationships in online persuasion, as they

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allow systematic manipulation of communication variables while controlling for potential confounding factors (Ha and Lennon, 2010; Liu and Shrum, 2009). Consistent with prior research on interactive advertising and website effectiveness, a between-subjects design was adopted to compare consumer responses across different levels of website interactivity. A between-subjects design is particularly suitable in this context, as it enables causal inference regarding the effects of interactivity levels while minimizing potential learning effects and demand characteristics across conditions.

The independent variable was the level of perceived interactivity embedded in a brand website, operationalized at three levels: low, medium, and high. The dependent variables included (1) thought elicitation related to the website and the product, (2) attitude strength, (3) attitudes toward the website and the brand, (4) intention to revisit the website, and (5) pre-purchase intention. This design enabled a comprehensive examination of cognitive, attitudinal, and behavioural responses, in line with ELM-based research in digital contexts (Sicilia *et al*., 2005; Oh and Sundar, 2015).

4.2 Product involvement and pre-tests

Given the central role of product involvement in the Elaboration Likelihood Model, particular attention was paid to the selection of an appropriate low-involvement product. Product involvement refers to the perceived personal relevance or importance of an object to an individual and encompasses both situational and enduring components (Antil, 1984; Zaichkowsky, 1986). Although involvement varies across individuals, prior research suggests that certain product categories tend to be perceived as low involvement by specific consumer groups (Rahtz and Moore, 1989).

Two pre-tests were conducted following established procedures (Hong, 2015; Han and Kim, 2017). In the first pre-test, participants were asked to identify product categories requiring minimal cognitive effort and personal relevance. In the second pre-test, involvement with the shortlisted products was measured using the Personal Involvement Inventory (PII; Zaichkowsky, 1986, 1994). Results indicated that a refreshment soft drink was consistently perceived as a low-involvement product, consistent with prior findings (Li *et al*., 2021). This product category was therefore selected for the experimental manipulation.

4.3 Development of experimental stimuli

Three versions of a fictitious brand website promoting a refreshment soft drink were developed, each corresponding to a different level of website interactivity (low, medium, and high). A fictitious brand was used to eliminate potential confounding effects related to prior brand familiarity, brand attitudes, or brand loyalty, thereby enhancing internal validity (Coyle and Thorson, 2001; Liu and Shrum, 2009).

All three website versions contained identical core content, including product descriptions, textual information, and visual imagery. This ensured that any observed differences in consumer responses could be attributed to variations in interactivity rather than differences in informational content or argument strength (Andrews and Shimp, 1990). The manipulation focused exclusively on the number and type of interactive features embedded in the website interface.

4.4 Manipulation of actual interactivity

Interactivity was conceptualised as a multidimensional construct comprising user control, two-way communication, and synchronicity (McMillan *et al*., 2003; Liu and Shrum, 2009). The manipulation of actual interactivity was guided by the framework proposed by Voorveld *et al*. (2011), which identifies website functions that enhance perceived interactivity.

Across the three website versions, interactive features included feedback forms, dropdown menus, content customization options, and tools enabling user-controlled navigation. The low-

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interactivity version featured minimal interactive elements and largely static navigation. The medium-interactivity version incorporated a moderate number of interactive features designed to facilitate engagement without excessive complexity. The high-interactivity version included a richer set of interactive elements intended to maximize user control, responsiveness, and perceived two-way communication.

To assess the validity of the manipulation, three experienced web-brand designers independently evaluated the website versions. All evaluators confirmed that the websites differed systematically in their level of actual interactivity and were consistent with the intended low, medium, and high interactivity conditions.

4.5 Measurement of perceived interactivity

Because perceived interactivity reflects users' psychological experience and is closely linked to engagement and persuasion, it was measured explicitly (Wu, 2006; Joo and Yang, 2023). Seventeen items were adapted from established scales developed by Liu (2003), McMillan and Hwang (2002), and Song and Zinkhan (2008), capturing the dimensions of active control, two-way communication, and synchronicity.

A pre-test involving 60 participants confirmed that perceived interactivity increased systematically from the low-to the medium- and high-interactivity website versions, supporting the effectiveness of the manipulation.

4.6 Sample and procedure

The sample consisted of 87 undergraduate students in their third or fourth year of study, recruited through convenience sampling. Students were considered appropriate participants due to their familiarity with digital environments and online shopping interfaces, consistent with prior experimental research (Sicilia *et al*., 2005; Liu and Shrum, 2009). The average age of participants was 20.9 years, and 48% were female.

Participants were randomly assigned to one of the three experimental conditions. Upon arrival at a university laboratory, participants completed an online questionnaire assessing demographic characteristics and general internet usage. They then navigated one version of the experimental website at their own pace. Following the navigation task, participants completed questionnaires measuring perceived interactivity, attitudes toward the website and the brand, and pre-purchase intentions.

To assess attitude strength, we follow the "operative measure" as proposed by Bassili (1996) and further analysed by Petty *et al*. (2023). To reduce immediate context effects, participants were contacted one week later and asked to complete a follow-up questionnaire without revisiting the website, consistent with prior research on delayed attitude measurement (Krosnick and Petty, 1995; DeMarree *et al*., 2007).

4.7 Measures

Thought elicitation was measured using a thought-listing technique, which captures cognitive responses during persuasion (Wright, 1973). Participants listed all thoughts that occurred while interacting with the website. Two trained coders independently categorized the thoughts as website-oriented or product-oriented. Discrepancies were discussed and resolved through consensus and, when necessary, by consulting a third coder, following established procedures in thought-listing research (Miniard *et al*., 1991).

Attitude strength was assessed using a multi-item scale adapted from Chen and Wells (1999) and operationalized following DeMarree *et al*. (2007). Attitudes toward the website and the brand were measured using semantic differential scales adapted from MacKenzie *et al*. (1986). Intention to revisit the website and pre-purchase intention were measured using scales developed by Kim and Biocca (1997) and applied in interactive advertising research (Coyle and Thorson, 2001).

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All measurement items were assessed using seven-point Likert-type scales. Perceived interactivity was measured through items capturing active control (e.g. "This website allows me to control my navigation"), two-way communication (e.g. "The website enables interaction between the user and the system"), and synchronicity (e.g. "The website responds quickly to my actions"). Attitudes toward the website and the brand were measured using semantic differential scales (e.g. bad/good, unappealing/appealing), while behavioural intentions were captured through items such as "I would revisit this website" and "I would consider purchasing this product online." All items were adapted from prior studies and modified to fit the context of the present research.

5. Results

5.1 Manipulation checks

Before testing the hypotheses, manipulation checks were conducted to verify whether the three experimental website versions differed as intended in terms of perceived interactivity. Perceived interactivity was measured using a multi-item scale capturing active control, two-way communication, and synchronicity.

A one-way analysis of variance (ANOVA) revealed a statistically significant effect of interactivity level on perceived interactivity, confirming that participants clearly distinguished between the low-, medium-, and high-interactivity website versions. Mean perceived interactivity increased monotonically across conditions, supporting the effectiveness of the experimental manipulation. Descriptive statistics and ANOVA results are summarized in (Table 1).

Effect sizes (partial eta squared, ηp^2) are reported to provide an estimate of the magnitude of the observed effects.

5.2 Thought elicitation (H1)

Hypothesis 1 examined whether the level of website interactivity influences the number of elicited thoughts related to (i) the website and (ii) the product.

A multivariate analysis of variance (MANOVA) with website-oriented and product-oriented thoughts as dependent variables revealed a statistically significant multivariate effect of interactivity level (Wilks' $\Lambda = 0.666$, $F = 9.349$, $p < 0.001$, $\eta p^2 = 0.18$), indicating that interactivity influenced cognitive responses overall.

Follow-up univariate ANOVAs showed that interactivity level had a significant effect on product-oriented thoughts ($F = 13.931$, $p < 0.001$, $\eta p^2 = 0.24$), whereas the effect on website-oriented thoughts was not statistically significant ($F = 3.568$, $p = 0.086$, $\eta p^2 = 0.08$). Descriptive statistics and ANOVA results are reported in Table 2.

Post-hoc Tukey HSD comparisons indicated that participants exposed to the medium-interactivity website generated significantly more product-related thoughts than those exposed to the low-interactivity website (Mean Difference = 3.09, $p < 0.001$). Similarly, participants in the high-interactivity condition generated significantly more product-related thoughts than those in the low-interactivity condition (Mean Difference = 2.43, $p < 0.001$). No significant

Table 1. Pre-test and descriptive statistics for perceived interactivity

Product involvement ANOVA	Low F	Sig.
Low vs high	2.978	0.011
Medium vs high	1.744	0.043
Medium vs low	1.745	0.045

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Table 2. Thought elicitation results (ANOVA and key post-hoc contrasts)

Dependent variable	F	df	*p*	ηp^2	Key contrasts
Website-oriented thoughts	3.5	2.87	0.086	0.08	n.s.
Product-oriented thoughts	13.93	2.87	<0.001	0.24	Medium > Low; High > Low

Note(s): Full Tukey HSD results are reported in the Online Supplementary Material
ANOVA results for website- and product-oriented thoughts and key significant Tukey HSD contrasts

difference emerged between the medium- and high-interactivity conditions. Detailed post-hoc comparisons are reported in the online supplementary material (Table S1). Full ANOVA tables are provided in the supplementary material (Tables S2-S5).

In contrast, none of the pairwise comparisons for website-oriented thoughts reached statistical significance, although descriptively higher means were observed for the medium-interactivity condition.

Overall, these findings partially support H1. Specifically, H1(ii)(a) and H1(ii)(c) are supported for product-oriented thoughts, whereas hypotheses related to website-oriented thoughts are not supported. The results indicate that, under low-involvement conditions, interactivity primarily influences cognitive processing related to the product rather than to the website itself.

5.3 Attitude strength (H2)

Hypothesis 2 proposed that attitude strength toward the low-involvement product would differ across levels of website interactivity. The attitude strength scale demonstrated satisfactory internal consistency (Cronbach's $\alpha = 0.779$).

A one-way ANOVA revealed a statistically significant effect of interactivity level on attitude strength ($F = 5.89$, $p = 0.003$, $\eta p^2 = 0.12$). Participants exposed to the high-interactivity website reported the strongest attitudes, followed by those in the low- and medium-interactivity conditions.

Post-hoc Tukey HSD comparisons indicated that attitude strength was significantly higher in the high-interactivity condition compared to the medium-interactivity condition (Mean Difference = 1.20, $p = 0.003$) and the low-interactivity condition (Mean Difference = 0.88, $p = 0.044$). The difference between the low- and medium-interactivity conditions was not statistically significant. These effects are illustrated in Figure 1.

Thus, H2 is partially supported: H2(b) and H2(c) are supported, whereas H2(a) is not supported. The findings suggest that attitude strength increases primarily when interactivity reaches a high level rather than through incremental increases from low to medium interactivity.

5.4 Attitudes toward the website and the brand (H3)

5.4.1 Attitude toward the website. Attitude toward the website was measured using a six-item scale with high internal reliability (Cronbach's $\alpha = 0.793$). A one-way ANOVA revealed a statistically significant effect of interactivity level on website attitudes ($F = 5.193$, $p = 0.007$, $\eta p^2 = 0.11$), as shown in Table 3.

Post-hoc Tukey HSD comparisons indicated that participants exposed to the low-interactivity website reported significantly more positive website attitudes than those exposed to the medium-interactivity website (Mean Difference = 0.49, $p = 0.007$). Differences involving the high-interactivity condition were not statistically significant. Full post-hoc results are reported in the supplementary material (Table S2).

These results suggest that, for low-involvement products, simpler website designs may be evaluated more favourably than moderately interactive ones. The pattern of means is illustrated in Figure S1 in the supplementary material.

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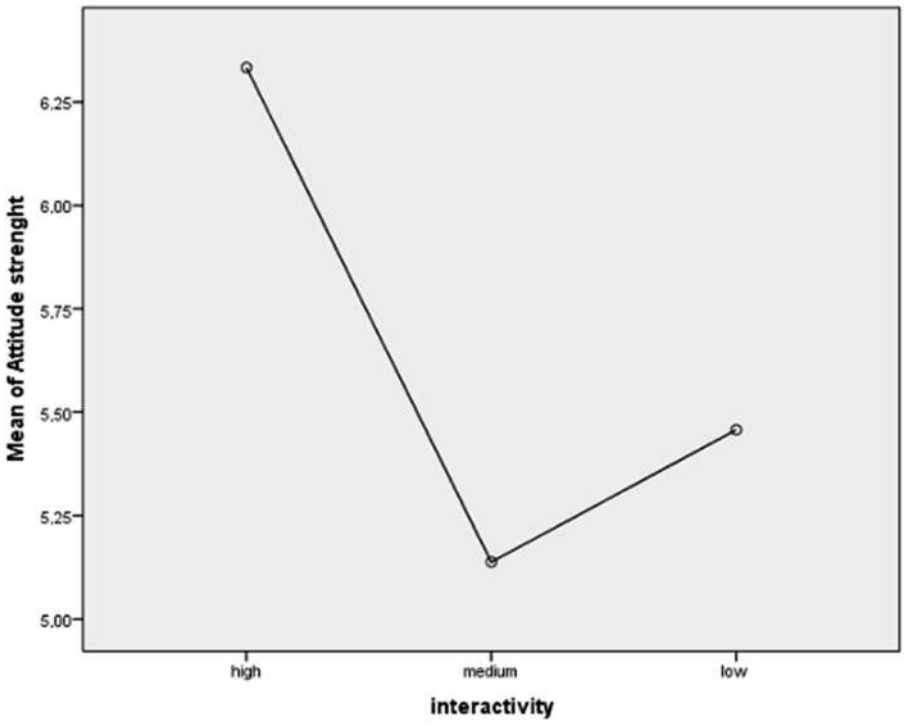


Figure 1. Attitude strength by interactivity level. Note: Attitude strength was highest in the high-interactivity condition

Table 3. Attitudinal outcomes (ANOVA summary)

Outcome	F	df	*p*	ηp^2	Result
Website attitude	5.19	2.87	0.007	0.11	Significant
Brand attitude	3.00	2.87	0.055	0.06	Marginal

Note(s): Pairwise comparisons are reported in the Supplementary Material

5.4.2 Attitude toward the brand. Brand attitude was assessed using a three-item scale with high reliability (Cronbach's $\alpha = 0.875$). Although mean brand attitude was highest in the high-interactivity condition, the overall ANOVA did not reach conventional levels of statistical significance ($F = 3.003$, $p = 0.055$, $\eta p^2 = 0.06$; Table 3).

Post-hoc analyses indicated marginal differences between the high- and medium-interactivity conditions, but no pairwise comparisons reached statistical significance. Detailed post-hoc comparisons are reported in the supplementary material (Table S3). Mean differences across interactivity conditions are illustrated in Figure S2 in the supplementary material.

Accordingly, H3(1) is not supported for brand attitude, although the observed directional pattern suggests a potential positive influence of high interactivity that does not reach statistical significance in low-involvement contexts.

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5.5 Behavioural intentions (H3)

Behavioural intentions were examined in terms of intention to return to the website and pre-purchase intention.

A one-way ANOVA revealed no statistically significant effects of interactivity level on intention to return to the website ($F = 1.642$, $p = 0.200$, $\eta p^2 = 0.04$) or purchase intention ($F = 1.749$, $p = 0.180$, $\eta p^2 = 0.04$). ANOVA summaries are presented in Table 4.

Post-hoc analyses confirmed that none of the pairwise comparisons reached statistical significance. Detailed Tukey HSD results are reported in the supplementary material (Tables S4 and S5). The patterns for intention to return and purchase intention are illustrated in Figures S3 and S4 in the supplementary material.

Overall, although higher levels of interactivity were directionally associated with stronger behavioural intentions, these effects were not statistically significant. The small effect sizes further indicate that the influence of interactivity on behavioural intentions is limited under low-involvement conditions. Consequently, H3 is only partially supported.

6. Discussion

The purpose of this study was to examine how different levels of website interactivity influence cognitive, attitudinal, and behavioural responses in the context of a low-involvement product, drawing on the ELM as the guiding theoretical framework. Overall, the findings demonstrate that website interactivity plays a meaningful but non-linear role in online persuasion. Rather than exerting uniformly positive effects, interactivity operates contingently, functioning primarily as a peripheral cue whose persuasive influence varies depending on its intensity and the type of consumer response examined.

6.1 Interactivity and cognitive processing under low involvement

Partially consistent with Hypothesis 1, interactivity significantly influenced product-oriented thoughts but not website-oriented thoughts. Participants exposed to medium and high levels of interactivity generated more product-related thoughts than those exposed to low interactivity, with the medium-interactivity condition producing the highest level of cognitive engagement.

This pattern aligns with the sixth (6th) postulate of ELM, which suggests that when both motivation and ability to process information are low, persuasion is driven primarily by peripheral cues rather than systematic argument evaluation (Petty and Wegener, 1998). In this context, interactivity appears to function as an experiential cue that enhances product salience without directing cognitive resources toward evaluation of the website itself. Rather than stimulating deep elaboration, interactivity may selectively increase attention to the product, reinforcing its role as a peripheral facilitator of persuasion.

The absence of significant differences in website-oriented thoughts further suggests that low-involvement consumers allocate limited cognitive resources to evaluating the medium. This finding supports prior research indicating that under low involvement, consumers rely on simplifying cues rather than engaging in comprehensive assessment of contextual elements (Petty and Cacioppo, 1986; Lee, 2020).

Table 4. Behavioral intentions (ANOVA summary)

Outcome	F	df	*p*	ηp^2	Result
Intention to return	1.64	2.87	0.200	0.04	Not significant
Purchase intention	1.75	2.87	0.180	0.04	Not significant

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6.2 Non-linear effects of interactivity and attitude strength

Findings related to attitude strength further underscore the non-linear nature of interactivity effects. Attitude strength was significantly higher in the high-interactivity condition compared to both the medium- and low-interactivity conditions, while no significant difference emerged between the low and medium levels. This pattern partially supports Hypothesis 2 and indicates that incremental increases in interactivity do not necessarily strengthen attitudes.

From an ELM perspective, this result can be interpreted through the third (3rd) postulate, which posits that the same variable may enhance or inhibit persuasion depending on contextual conditions (Petty and Cacioppo, 1986). Moderate interactivity may attract attention without providing sufficient experiential reinforcement to strengthen attitudes meaningfully. In contrast, high interactivity may reach a threshold at which peripheral cues - such as enjoyment, perceived control, and immersion - enhance attitude accessibility and confidence, contributing to stronger attitudes even in the absence of extensive elaboration.

This interpretation is consistent with prior research suggesting that interactivity enhances affective involvement and experiential value, thereby reinforcing attitude strength (Flavián *et al*., 2019; Yang and Shen, 2018). At the same time, the absence of a linear progression highlights the importance of identifying optimal rather than maximal interactivity, particularly in low-involvement contexts.

6.3 Website and brand attitudes: the role of simplicity

A theoretically meaningful finding emerged with respect to attitudes toward the website. Participants exposed to the low-interactivity website reported more positive website attitudes than those exposed to the medium-interactivity version. This result suggests that under low-involvement conditions, simpler website designs may be evaluated more favourably than moderately interactive ones.

This pattern aligns with research emphasizing the cognitive costs associated with interactivity. When motivation to process information is low, even moderate levels of interactivity may introduce unnecessary complexity, increasing perceived effort and reducing evaluative responses (Ariely, 2000; Fortin and Dholakia, 2005). In such cases, low interactivity may function as a heuristic cue signalling ease of use and efficiency, thereby enhancing website evaluations.

Brand attitudes, by contrast, exhibited only marginal and non-significant differences across interactivity levels. Although evaluations were directionally more positive in the high-interactivity condition, these effects were weak, suggesting that brand-related responses remain relatively insensitive to contextual cues under low involvement. This finding is consistent with prior research indicating that brand attitudes formed under low involvement are less adaptable and may require repeated exposure or higher involvement to strengthen meaningfully (Pangarkar and Kirk, 2025).

6.4 Interactivity and behavioural intentions

Results related to behavioural intentions further support the contingent role of interactivity in online persuasion. Although higher interactivity was associated with descriptively higher intentions to revisit the website and to purchase the product, these differences were not statistically significant. This finding indicates that positive experiential responses generated by interactivity do not automatically translate into behavioural intentions for low-involvement products.

From an ELM perspective, this outcome is theoretically coherent. Peripheral cues may shape affective evaluations and attitude strength without producing durable behavioural outcomes, particularly when motivation to act remains limited (Krosnick and Petty, 1995). Behavioural intentions may require stronger central processing, higher involvement, or repeated exposure to reinforce the translation of positive attitudes into action. This finding is consistent with broader evidence suggesting that the relationship between attitudes and

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behavioural outcomes is often indirect and mediated by additional factors such as engagement and brand-related experiences (Desveaud *et al*., 2024).

Consistent with prior research, the findings suggest that interactivity can enhance engagement and enjoyment without yielding proportional increases in purchase-related intentions in low-involvement contexts (Liu and Shrum, 2009; Islam *et al*., 2021; Shabani Shojaei and Barbosa, 2026). Collectively, these results caution against assuming a direct or linear relationship between website interactivity and behavioural outcomes.

7. Implications

7.1 Theoretical implications

This study offers several theoretical contributions to the literature on online persuasion, website interactivity, and consumer behaviour by extending the ELM to contemporary digital environments characterized by varying degrees of interactivity. By focusing explicitly on a low-involvement product context, the study advances a contingent understanding of interactivity effects, challenging assumptions of uniformly positive outcomes.

First, the findings provide empirical support for the view that website interactivity operates primarily as a peripheral cue under low-involvement conditions. By demonstrating that interactivity influences product-related thought elicitation and attitude strength without consistently enhancing behavioural intentions, the study substantiates ELM's sixth (6th) postulate, which emphasizes the dominance of peripheral cues when motivation and ability to process information are low (Petty and Wegener, 1998). In doing so, the study clarifies how interactivity contributes to persuasion without necessarily triggering central-route processing.

Second, the results highlight the non-linear effects of interactivity. Different levels of interactivity exert distinct effects on cognitive and attitudinal outcomes: medium interactivity maximizes product-related thought elicitation, whereas high interactivity is required to strengthen attitudes meaningfully. This pattern aligns with ELM's third (3rd) postulate, which suggests that the same variable may facilitate or inhibit persuasion depending on contextual conditions (Petty and Cacioppo, 1986). By empirically demonstrating threshold effects, the study moves beyond linear conceptualisations and supports calls for level-specific theorizing in interactivity research (Yang and Shen, 2018; Maduku *et al*., 2024).

Third, the study contributes to the interactivity literature by distinguishing between evaluations of the website and evaluations of the brand. The finding that low interactivity yields more favourable website attitudes than medium interactivity challenges the assumption that increasing interactivity uniformly enhances user evaluations. In low-involvement contexts, simplicity and ease of use appear to function as powerful peripheral cues shaping evaluations of the communication medium. By contrast, brand attitudes are less sensitive to interactivity variations, reinforcing the notion that brand-related evaluations under low involvement may require repeated exposure or stronger affective reinforcement to shift meaningfully.

Finally, by explicitly integrating product involvement into the analysis, the study reinforces its role as a critical boundary condition in online persuasion. The findings demonstrate how low involvement constrains the persuasive impact of interactivity across cognitive, attitudinal, and behavioural outcomes, contributing to a more conditional and context-sensitive understanding of digital persuasion mechanisms.

7.2 Managerial implications

Beyond theory, the findings offer several practical implications for managers designing digital marketing strategies and brand websites.

First, the results caution against the assumption that more interactivity is always better. For low-involvement products, increasing interactivity does not necessarily lead to proportional gains in engagement or behavioural intentions. Managers should therefore avoid overloading websites with interactive features that increase cognitive effort without delivering clear value.

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Second, the finding that low-interactivity websites generate more positive website attitudes highlights the importance of simplicity and usability. In low-involvement contexts, consumers may prioritize efficiency, clarity, and ease of navigation over rich interactive experiences. Streamlined website designs that minimize friction may therefore be more effective. For example, firms marketing beverages or other FMCG products may benefit more from fast-loading pages, intuitive navigation, and clear product presentation than from adding multiple interactive tools that may increase cognitive effort.

Third, high levels of interactivity appear to enhance attitude strength, suggesting that immersive experiences may still be valuable when the strategic objective is to build favourable attitudes rather than to drive immediate purchase behaviour. Interactive elements that foster enjoyment, perceived control, or engagement may strengthen affective associations with the product or brand.

Finally, the absence of significant effects on behavioural intentions underscores the need for integrated digital strategies. Interactivity alone may be insufficient to drive action in low-involvement contexts. Managers should combine interactive website features with complementary tactics, such as promotions, reminders, or repeated exposure across channels, to reinforce behavioural responses over time.

8. Limitations and future research

Despite its contributions, several limitations should be acknowledged, which also suggest directions for future research.

First, the study employed an experimental design using a student sample, which may limit generalizability. Although students are appropriate participants for online persuasion research due to their digital familiarity, future studies should replicate the design using more heterogeneous samples to assess the robustness of the findings across demographic groups.

Second, the study focused on a single low-involvement product category. While this choice was theoretically justified and empirically validated, involvement is context-dependent. Future research could examine whether the observed non-linear effects extend to other low-involvement categories or compare low- and high-involvement products within the same framework.

Third, interactivity was operationalized through a specific set of website features. As digital technologies evolve, future research should explore whether emerging forms of interactivity - such as AI-driven personalization, conversational agents, or immersive technologies - produce similar or distinct persuasive effects.

Fourth, the study relied on behavioural intentions rather than actual behaviour. Although intentions are theoretically meaningful proxies, future research could enhance external validity by incorporating behavioural measures, such as click-through rates or real purchase data.

Finally, while attitude strength was measured with a temporal delay, the overall time frame remains limited. Longitudinal research could examine whether interactivity-induced attitudes persist, decay, or strengthen over time, providing deeper insight into the durability of peripheral-route persuasion in low-involvement contexts.

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Supplementary material

The supplementary material for this article can be found online.

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